

Agniveer Vayu Other than Science (Group Y) - 22 March 2025 - Memory Based Paper

English

Question 1

Change the given statement to indirect speech.

Tom: “I’ll be there at eight. Is that all right?”

Options:

- A. Tom said that he would be there at eight and he checked if it was all right.
- B. Tom said that he will be there at eight and asked if it was all right.
- C. Tom told that he would be there at eight. And asked if it is all right.
- D. Tom said he will there at eight and asked if that is all right.

Answer: A

Solution:

The correct answer is 'Tom said that he would be there at eight and he checked if it was all right'.

Key Points

- While changing the given sentence from **direct to indirect**, we need to follow the given steps -
- The sentence is in **simple future tense**, so here '**will**' changes to '**would**' in indirect speech.
- The pronoun will change according to the subject of the reporting verb. So, here '**I**' to '**he**'. (the verb is not mentioned in the given sentence)
- We will use the reporting verb '**said**' followed by '**that**'.
- We will use '**and**' to connect the two sentences within the reported speech.
- Since the second sentence in the reported speech starts with **an interrogative verb**, we will use 'if' in indirect speech.
- Thus, based on the given points, the correct answer is **Option 1**.

Correct sentence is indirect speech - *Tom said that he would be there at eight and he checked if it was all right.*

Question 2

A sentence has been given with a blank to be filled with an appropriate option. Choose the correct alternative.

John rushed to the spot and saw his wife and kid _____ on the road.

Options:

A. lain

B. lying

C. lie

D. laying

Answer: B

Solution:

The correct answer is 'Option 2 i.e. (lying)'



Key Points

- The correct verb to use in this context is "lying" as it refers to being in or assuming a horizontal or resting position on a surface.
- "Lying" is the present participle of the verb "to lie."
- Therefore, the sentence should read, "John rushed to the spot and saw his wife and kid lying on the road."
- This makes Option 2 (lying) the correct choice to fill the blank.

Therefore, the correct sentence is: "John rushed to the spot and saw his wife and kid lying on the road."



Additional Information

- "lain" - This is the past participle form of the verb "to lie" but is not suitable in this context.
- "lie" - This is the base form of the verb, not correct in this sentence's tense structure.
- "laying" - This form of the verb typically requires an object, e.g., 'laying the blanket on the bed', and is not correct here.

Question 3

Choose the part of the sentence that is incorrect in questions.

He said (1)/ that the horse (2)/ died at (3)/ the night.(4)

Options:

A. (1)

B. (2)

C. (3)

D. (4)

Answer: C

Solution:

The correct answer is '3 - died at.'



Key Points

- The phrase "**died at the night**" is incorrect. The correct preposition to use with "night" is "**in**," not "at."
- "**At**" can be used with specific times, like "**at 10 o'clock**," but with periods of the day like "night," the correct preposition is "**in**."
- Therefore, the correct sentence should read: "**He said that the horse died in the night.**"
- The rest of the sentence is grammatically correct and follows the proper tense and structure for indirect speech.

Therefore, the correct answer is Option 3.

Question 4

In the following questions, out of the four alternatives choose the one which can be substituted for the given words / sentence.

'Edible' means that which is:

Options:

A. Worth eating

B. Can not be deleted

C. Can be edited

D. Edifying

Answer: A

Solution:

The correct answer is option 1 i.e. 'Worth eating'.

 Key Points

- **Edible:** (खाद्य) fit to be eaten.
 - **Example:** "The fruits and vegetables in the market were all edible."
- **Worth eating:** This is a synonym for edible.

Therefore, the correct answer is - **Worth eating**

 Additional Information

- **Can not be deleted:** This refers to something that cannot be removed or erased.
 - **Can be edited:** This refers to something that can be changed or modified.
 - **Edifying:** This refers to something that is morally or intellectually uplifting.
-

Question 5

Select the word **OPPOSITE** in meaning to the given word.

Deficient

Options:

A. Skimpy

B. Broken

C. Ample

D. Terrible

Answer: C

Solution:

The correct answer is: **Option 3.**

 Key Points

- The word "**Deficient**" means lacking in some necessary quality or element; not enough in amount or degree. (अपर्याप्त)
 - **Example:** The diet was deficient in vitamins, leading to health problems.
- "**Ample**" refers to being more than enough in quantity, size, or scope; fully sufficient or more than adequate. (प्रचुर)
 - **Example:** There was ample space in the room for all the guests to sit comfortably.
- Hence, we can infer that the opposite of '**Deficient**' is '**Ample**'.

Therefore, the correct answer is: **Option 3**.

Additional Information

Here are the other options explained along with their Hindi meanings and example sentences:

- **Skimpy** (अल्प): Lacking in size, fullness, or quantity; meager.
 - **Example:** The portions of food served were quite skimpy, leaving everyone still hungry.
- **Broken** (टूटा हुआ): Separated into parts or pieces due to being damaged.
 - **Example:** The broken vase lay in fragments on the floor.
- **Terrible** (भयानक): Extremely or distressingly bad or serious.
 - **Example:** The weather was so terrible that we decided to stay indoors all day.

Question 6

Find the part of the given sentence that has an error in it. If there is no error, choose 'No error'.

It is believed that difficult tasks demand lot of our resources.

Options:

- A. It is believed that
- B. No error
- C. difficult tasks demand lot
- D. of our resources.

Answer: C

Solution:

The correct answer is '**Option 3**'

 **Key Points**

- The error in the sentence is in the segment **"difficult tasks demand lot."**
- The word **"lot"** should be preceded by the article **"a"** to make the phrase grammatically correct.
- The corrected part of the sentence should be **"difficult tasks demand a lot."**

Therefore, the **correct sentence** is: *"It is believed that difficult tasks demand a lot of our resources."*

Additional Information

- **Option 1:** "It is believed that" is grammatically correct.
 - **Option 4:** "of our resources" is grammatically correct.
-

Question 7

Some parts of a sentence have been jumbled up, and labelled P, Q, R and S. Select the option that gives the correct sequence in which these parts can be rearranged to form a meaningful and grammatically correct sentence.

The Government of India Act was passed by

P. separate acts namely, the Government of India Act 1935 and

Q. the longest act enacted by the British Parliament

R. the British Parliament in August 1935, it was

S. at that time, hence it was divided into two

the Government of Burma Act 1935,

Options:

A. QSPR

B. PQRS

C. SPQR

D. RQSP

Answer: D

Solution:

The correct answer is 'RQSP'.



Key Points

- The sentence begins with the main subject "The Government of India Act", which was passed by "the British Parliament in August 1935", indicating the starting point as segment R.
- The next logical information is the fact that it was "the longest act enacted by the British Parliament", which is given in segment Q.
- Due to its length, it was divided into two separate acts, as described in segments S and P.
- Thus, the complete sentence should read: "The Government of India Act was passed by the British Parliament in August 1935, it was the longest act enacted by the British Parliament at that time, hence it was divided into two separate acts namely, the Government of India Act 1935 and the Government of Burma Act 1935."

Therefore, the correct answer is Option 4.

Question 8

A sentence has been given with a blank to be filled with an appropriate option. Choose the correct alternative.

I _____ the cake before my mother arrived.

Options:

- A. had baked
- B. bake
- C. baked
- D. have baked

Answer: A

Solution:

The correct answer is 'Option 1' i.e. "had baked."



Key Points

- In this sentence, two past events are mentioned: baking the cake and mother's arrival.

- When **two actions occur** in the **past**, the action that happened first is usually in the past perfect tense, while the action that happened second is in the simple past tense.
- In this sentence, "*before my mother arrived*" is in the **simple past**, so the baking of the cake, which happened first, should be in the past perfect ("had baked").

Therefore, the **correct sentence** is: '*I **had baked** the cake before my mother arrived.*'

Additional Information

- Here's the explanation for each option:
 - **"Bake"** - This is the base form of the verb, typically used in the present tense. It is not suitable for this sentence, which refers to past actions.
 - **"Baked"** - This is the past simple form of the verb. It can be used to indicate a completed action in the past, but when describing two events that both occurred in the past, the action that happened first (baking the cake) should be in the past perfect tense.
 - **"Have baked"** - This is the present perfect form of the verb. It's used to connect the past and the present, often to describe an action that happened in the past but is still relevant in the present. It does not fit in this sentence, which refers to two distinct past events.
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Question 9

Choose the word opposite in meaning to the given word.

UNSCRUPULOUS

Options:

- A. Dedicated
- B. Single-minded
- C. Superfluous
- D. Conscientious

Answer: D

Solution:

The correct solution is **option 4 i.e. Conscientious**.



Explanation

Let us know the meaning of the words given:

- **Dedicated** means to be devoted to a task or purpose.
- **Single-minded** means concentrating on only one aim.
- **Superfluous** means unnecessary, especially through being more than enough.
- **Conscientious** means wishing to do one's work or duty well and thoroughly.

The given word '**Unscrupulous**' means having or showing no moral principles; not honest or fair. So, the word opposite in meaning to the given word is '**Conscientious**'.

Question 10

Here is a letter to MTNL complaining that your phone has not been working for a few days. The sentences have got jumbled up. Rearrange the sentences to make the complaint coherent.

I. My landline phone has not been working for a week.

II. I am a resident of House No. X, Street Y, Chennai.

III. Please set my phone right at the earliest.

IV. I informed the Area Manager telephonically about it but there is no response.

Options:

A. I, II, III, IV

B. II, I, IV, III

C. IV, III, II, I

D. III, II, IV, I

Answer: B

Solution:

The correct answer is '**II, I, IV, III**'.



Key Points

- The **first** sentence will introduce the sender and his/her address. Hence, '**II**' will be the first sentence of the letter.
- The **second** sentence will explain the problem. Hence, '**I**' will be the second sentence of the letter.
- The next sentence will be '**IV**' as it states the sender's efforts of contacting the **Area Manager**.
- The last statement will be **III**, which is **requesting** to repair the phone as early as possible.

Hence, the correct answer is **Option 2**.

Question 11

Four words are given out of which only one word is correctly spelt. Find the correctly spelt word.

Options:

- A. Greatfull
- B. Greatful
- C. Gratefull
- D. Grateful

Answer: D

Solution:

The correct answer is "**Grateful**".



Key Points

- "**Grateful**" is the correct spelling of the word.
- **Grateful** - It means **feeling or showing appreciation for something received or done**.
- When you are grateful, you are thankful for the kindness, help, or opportunities that you have received.
 - **For Example:** "I am truly **grateful** for all the support you've given me during my difficult times."

Therefore, the correct answer is "**Option 4**".

Question 12

Direction: Select the most appropriate option to fill in the blank.

I enjoy finding ways to make the most of my budget by being _____.

Options:

- A. lethargic
- B. miser
- C. frugal
- D. gloomy

Answer: C

Solution:

The correct answer is 'Option 3' i.e. 'Frugal'.



Key Points

- The **sentence is about** maximizing the use of one's budget.
- The **blank should be filled** with a word that denotes careful or sparing use of money.
- "**Frugal**" (मितव्ययी) means economical in use or expenditure; not wasteful.

Therefore, the **correct sentence** is: "*I enjoy finding ways to make the most of my budget by being frugal.*"



Additional Information

- "**Lethargic**" (आलसी) means sluggish and lacking in energy, and does not relate to the context.
- "**Miser**" (कंजूस) refers to a person who hoards money and avoids spending, but this term can have a negative connotation.
- "**Gloomy**" (उदास) means dark or dim; deeply shaded, and is not related to the context of the sentence.

Question 13

Choose the option that is the active form of the sentence.

The roads have been dug up again by the municipal authorities.

Options:

- A. The municipal authorities have dug up the roads again.
- B. The municipal authorities are being dug up the roads again.

C. The municipal authorities are dug up the roads again.

D. The municipal authorities were dug up the roads again.

Answer: A

Solution:

The correct answer is "The municipal authorities have dug up the roads again".

Key Points

- In order to convert the given sentence into an active form, we need to identify the **tense** first of all.
- The given sentence is in the **present perfect tense**.
- The rule of the present perfect tense (active voice)-**Has/Have + Verb 3rd form**.
- The subject is '**the municipal authorities**' and the object is '**the roads**'.
- So the sentence would be -*The municipal authorities have dug up the roads again.*

Hence, the correct answer is **Option 1**.

Additional Information

- The criminal has been punished by the jailer.
 - *The jailer has punished the criminal.*
- The clothes have been washed by my friend.
 - *My friend has washed the clothes.*

Hinglish

- दिए गए वाक्य को सक्रिय रूप में बदलने के लिए, हमें सबसे पहले **tense** की पहचान करनी होगी।
- दिया गया वाक्य **present perfect tense** में है।
- **Present perfect tense** (active voice) का नियम **Has/Have + Verb 3rd form** है।
- The subject is '**the municipal authorities**' तथा object is '**the roads**'.
- तो वाक्य होगा -*The municipal authorities have dug up the roads again.*

अतः सही उत्तर **विकल्प 1** है।

Question 14

Which of the following is the correct noun form of the word 'Destroy' ?

Options:

A. Destructive

B. Destruction

C. Destruct

D. Destructing

Answer: B

Solution:

The correct answer is '**Option 2**', destruction.



Key Points

- The correct noun form of the word 'Destroy' is Destruction.
- Destructive is an adjective.
- Destruct is not commonly used as a noun.
- Destructing is a present participle form of the verb.

Hence, the correct answer is **option 2**.

Correct sentence: *The correct noun form of the word 'Destroy' is Destruction.*



Additional Information

- The noun 'Destruction' refers to the action or process of causing so much damage to something that it no longer exists or cannot be repaired.
- Examples of usage:
 - The hurricane caused widespread **destruction**.
 - The war led to the **destruction** of many historic buildings.

Question 15

Select the option that expresses the given sentence in passive voice.

His parents will send him to a boarding school next year

Options:

A. He will be sent to a boarding school next year by his parents.

B. He will be sent to a boarding school by his parents next year.

C. He is to be sent to a boarding school next year by his parents.

D. He would be sent to a boarding school next year by his parents.

Answer: A

Solution:

The correct answer is: "**He will be sent to a boarding school next year by his parents.**"

Key Points

- In passive voice, the subject of the active sentence becomes the receiver of the action, and the agent (the doer of the action) is either mentioned after the verb with the preposition "by" or omitted if it's not important to the sentence's meaning.
- The given sentence is in active voice with the subject "**His parents**" performing the action of sending "**him**" to a boarding school.
- To convert it into passive voice, we need to make the object "**him**" the subject of the sentence and add the helping verb "**will be**" followed by the past participle of the main verb "**sent**".
- In the passive sentence, "**He will be sent**" is the passive construction of "**His parents will send**," and "**by his parents**" indicates who is performing the action in a passive voice sentence.
- The preposition "**by**" is used to indicate the doer of the action, which is "his parents" in this case.

Therefore, the correct answer is **option 1**.

Question 16

Select the most appropriate synonym of the given word.

Grandeur

Options:

A. Magnificence

B. Liable

C. Commotion

D. Power

Answer: A

Solution:

The correct answer is '**Option 1**' i.e. '**Magnificence**'.

Key Points

- The word '**Grandeur**' typically means the quality of being grand or magnificent. (भव्यता).
 - **Example:** The grandeur of the royal palace left all the visitors in awe.
- The synonym '**Magnificence**' signifies the quality of being impressively beautiful or grand. (अति उत्कृष्टता).
 - **Example:** The magnificence of the natural landscapes is truly breathtaking.

Therefore, the **correct answer** is '**Magnificence**'.

Additional Information

- '**Liab**' as per Option 2 is an adjective that means responsible by law or likely to do or to be something. (उत्तरदायी).
 - '**Commo**' as per Option 3 is a noun that suggests a state of confused and noisy disturbance. (हलचल).
 - '**Power**' as per Option 4 typically refers to the ability to do something or act in a particular way, or the capacity to influence the behavior of others. (शक्ति).
-

Question 17

Read the passage given below and answer the questions that follow.

Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

The sweet almond is cultivated extensively in certain favourable regions, though nut crops are uncertain wherever frosts are likely to occur during flowering. While more than 25 types of almonds are grown in California, Marcona and Valencia almonds come from Spain, and Ferragnes are imported from Greece.

In which months, does the Almond tree produce flower?

Options:

- A. Late January to early April
- B. Late April to early January
- C. January
- D. March to May

Answer: A

Solution:

The correct answer is "**Late January to early April**"

Key Points

- You can find the highlighted lines in the **1st** paragraph:
 - Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from **late January to early April north** of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

Hence, the correct answer: *Late January to early April*

Question 18

Read the passage given below and answer the questions that follow.

Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

The sweet almond is cultivated extensively in certain favourable regions, though nut crops are uncertain wherever frosts are likely to occur during flowering. While more than 25 types of almonds are grown in California, Marcona and Valencia almonds come from Spain, and Ferragnes are imported from Greece.

Which of the following options is correct?

Options:

- A. The flowers of Almond have five petals.
- B. The flowers are self-incompatible
- C. The flowers require insect pollinators to facilitate cross-pollination
- D. All of the above

Answer: D

Solution:

The correct answer is "**All of the above**"

Key Points

- You can find the highlighted lines in the **1st** paragraph:
 - Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, **five-petaled**, light pink to white flowers from late January to early April north of the Equator. **The flowers are self-incompatible** and thus **require insect pollinators to facilitate cross-pollination with other cultivars**. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

Hence, the correct answer: *All of the above*

Question 19

Read the passage given below and answer the questions that follow.

Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

The sweet almond is cultivated extensively in certain favourable regions, though nut crops are uncertain wherever frosts are likely to occur during flowering. While more than 25 types of almonds are grown in California, Marcona and Valencia almonds come from Spain, and Ferragnes are imported from Greece.

What can be deduced from the given passage?

Options:

- A. Almond trees absorb a lot of water
- B. Almonds are not true nuts
- C. Almonds are exported from India to America
- D. None of the above

Answer: B

Solution:

The correct answer is "Almonds are not true nuts"

Key Points

- You can find the highlighted lines in the **1st** paragraph:
 - Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, **almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.**

Hence, the correct answer: *Almonds are not true nuts*

Question 20

Read the passage given below and answer the questions that follow.

Almond trees are deciduous with a hardy dormancy. Typically growing 3–4.5 metres (10–15 feet) tall, the trees are strikingly beautiful when in flower; they produce fragrant, five-petaled, light pink to white flowers from late January to early April north of the Equator. The flowers are self-incompatible and thus require insect pollinators to facilitate cross-pollination with other cultivars. The growing fruit (a drupe) resembles a peach until it approaches maturity; as it ripens, the leathery outer covering, or hull, splits open, curls outward, and discharges the pit. Despite their common label, almonds are not true nuts (a type of dry fruit) but rather seeds enclosed in a hard fruit covering.

The sweet almond is cultivated extensively in certain favourable regions, though nut crops are uncertain wherever frosts are likely to occur during flowering. While more than 25 types of almonds are grown in California, Marcona and Valencia almonds come from Spain, and Ferragnes are imported from Greece.

What does the word 'extensively' mean in the given passage?

Options:

- A. in a way that covers or affects a large area.
- B. in a way that covers or affects a tiny area.
- C. found in a particular region
- D. covering a sandy area

Answer: A

Solution:

The correct answer is "**in a way that covers or affects a large area.**"

Key Points

- Let's look at the meaning of the given word:

- **Extensively**-in a way that covers or affects a large area.
- Look at the example:
 - The house was **extensively** damaged by the fire

Hence, the correct answer: *in a way that covers or affects a large area*

Hinglish

- **Extensively**-बगीचे का एक ऐसा छेत्र जहाँ पर छोटे छोटे पौधे लगे होते है

RAGA

Question 21

Who is the author of the famous book 'Meghdoot'?

Options:

- A. Bhasa
- B. Kalidasa
- C. Sudraka
- D. Vatsyayana

Answer: B

Solution:

The correct answer is **option 2**, i.e. **Kalidasa**.

Writer Name	Books
Bhasa	Swapna vasavadatta
Kalidasa	Meghdoot
Sudraka	Mrchchhakatika
Vatsyayana	Kama Sutra

Question 22

Which of the following sports have the largest number of participants in terms of either side?

Options:

A.

Rugby football

B.

Water polo

C.

Baseball

D.

Cricket

Answer: A

Solution:

The correct answer is Rugby football.

Key Points

- Rugby is a collective name for the family of team sports of rugby union and rugby league, as well as the earlier forms of football from which both games, as well as Australian rules football and gridiron football, evolved.
- In 1820 the game of Rugby was played rather like soccer, but players were allowed to catch the ball and kick it out of their hands.
- **Rugby football has the largest number of participants in terms of either side. Hence, Option 1 is correct.**

Game	Number of Players
Cricket	11
Football	11
Hockey	11
Basketball	05
Baseball	09
Rugby Football	15

Game	Number of Players
Polo	04
Water Polo	07
Volleyball	06
Kabaddi	07
Lacrosse	12
Handball	07
Kho Kho	09
Sepak Takraw	03

Question 23

Why is tungsten used in the incandescent lamp filaments?

Options:

- A. Tungsten has high melting point
- B. Tungsten is less corroding at high temperatures
- C. Tungsten is very hard & difficult to break the thin filament
- D. Tungsten is an inert element in last column Group 18 periodic table

Answer: A

Solution:

CONCEPT:

Tungsten:

- The metal tungsten is used for the filaments in incandescent bulbs.
- It has a high melting point and retains its strength when heated.
- Filaments of the light bulbs are made up of the Tungsten element.
- Its symbol is 'W' because of its scientific name 'Wolfram' and its atomic number 74.

EXPLANATION:

- The materials commonly used as filament for incandescent lamps are carbon, tantalum, tungsten, and osmium.

- As the **resistance is less**, **heat energy** is produced is **very low** which is not sufficient for an **electric bulb to glow** so the **resistance is kept high**.
- **Tungsten** is very resistant to corrosion and has the **highest melting point** (melting point = 3695 K) and the highest tensile strength of any element.
- **Tungsten** is used for making bulb filaments of incandescent lamps because it has the **highest melting point and does not melt even while it is glowing for long hours**. Therefore option 1 is correct.

Additional Information

Properties of Tungsten:

- **The working temperature of Tungsten is 2,500-3,000°C.**
- Its resistance at working temperature is about 12-15 times the cold resistance.
- It has a positive temperature coefficient of resistance of 0.0045.
- Its resistivity is 12.5 $\mu\Omega$ -cm.
- Its density is 19.3
- The efficiency when working at 2,000°C is 18 lumens/W.

Question 24

In Microsoft Word, a collection of frequently used commands such as save, undo, print preview is located on the _____.

Options:

- A. Quick access toolbar
- B. Task bar
- C. Status bar
- D. Context Menu

Answer: A

Solution:

Correct answer is **Quick access toolbar**.

Key Points

- The **Quick Access Toolbar** provides access to **frequently used commands**, and the option to **customize the toolbar** with the commands that you use most often.
- By default, the **New, Open, Save, Quick Print, Run, Cut, Copy, Paste, Undo, and Redo buttons** appear on the **Quick Access Toolbar**.

- In Microsoft Word, a collection of frequently used commands such as **save, undo, and print preview** is located on the **Quick access toolbar**.

Additional Information

- **The taskbar** is an element of an **operating system** located at the **bottom of the screen**.
 - A **status bar** is a **graphical control element** used to **display certain status information** depending on the **application or device**.
 - A **context menu** is a **pop-up menu** that provides shortcuts for **actions the software developer anticipates** the user might want to take.
-

Question 25

Which is the most abundant metal in the Earth crust?

Options:

A. Copper

B. Iron

C. Aluminium

D. Silver

Answer: C

Solution:

- **Aluminium is the most abundant metal in the earth's crust.**
 - Around 8.1 % of the the total metals in the earth's surface is Aluminium.
 - The most abundantnon-metal in the earth's surface is Oxygen.
 - The Most abundant metalloid in the earth's crust is Silicon.
 - O > Si > Al > Fe > Ca are the most abundant element in earth crust.
-

Question 26

Which colour ribbon is used as a symbol for solidarity with HIV positive people and those living with AIDS?

Options:

- A. Blue
- B. Pink
- C. Red
- D. Orange

Answer: C

Solution:

The correct answer is '**Red**'.

Key Points

- **Acquired immunodeficiency syndrome (AIDS)** is a **chronic**, potentially **life-threatening condition** caused by the **human immunodeficiency virus (HIV)**.
- The **red ribbon** has become an **internationally recognized symbol** for **AIDS awareness**.
- It is used as a **universal symbol for solidarity with HIV-positive people and those living with AIDS**.
- It is worn by people in **remembrance of those who have died**.
- **World AIDS Day** is observed on **1 December** every year.

Important Points

Ribbon	Symbol
Blue	Child Abuse and Prevention
Pink	Breast Cancer
Orange	Leukemia, Kidney Cancer

Question 27

Time of zero-hour in Parliament is:

Options:

- A. 9 am to 10 am
- B. 10 am to 11 am
- C. 11 am to 12 noon
- D. 12 noon to 1 pm

Answer: D

Solution:

The correct answer is **12 noon to 1pm**.

- **Zero Hour** is an idea derived from the Indian Parliament itself.

Key Points

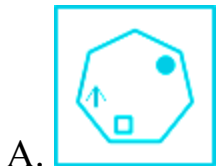
- It started in India. However, there is no mention of **zero hour** in the process of **Parliament proceedings**.
- The idea of '**Zero Hour**' came out in the first decade of **Parliament** when the MPs felt the need to raise the **national and important issues** of their parliamentary constituency.
- It is also said that this name was **given by newspapers**.
- In the early days there used to be a **lunch break in the Parliament at one o'clock**.
- In such a situation, **MPs got a good opportunity to raise the national issue at 12 noon** without any prior notice.
- During this time, they used to get an hour-long time.
- **Gradually this hour became to be known as 'Zero Hour'**.
- The presiding officers of both the Houses started directing to broadcast the proceedings of Zero Hour so that it could be made more effective.

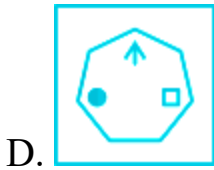
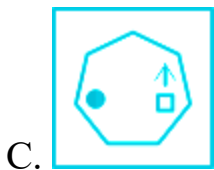
Question 28

Which of the given shapes would complete the sequence?



Options:





Answer: C

Solution:

The figure that will replace the question mark (?) will follow the following:

The point moves 2 corners anticlockwise-



The cube moves 2 corners clockwise-



The arrow moves 2 corners anticlockwise-



Therefore, the resultant series will be:



Hence, the correct answer is **option (3)**.

Question 29

Select the figure from among the given options that can replace the question mark (?) in the following series.



Options:

A.



B.



C.



D.



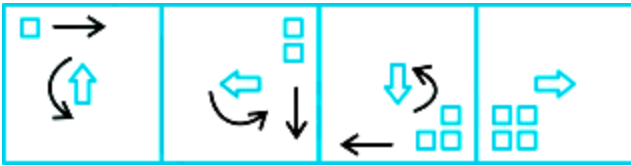
Answer: C

Solution:

The pattern followed here is:

1) The arrow is rotate 90° anticlockwise in the every next figure.

2) Square is shift their position form right upper corner in second figure, then right bottom corner in third figure at last left bottom corner in fourth figure and increase the number square in the next figure.



Hence, the correct answer is "**Option 3**".

Question 30

Four numbers have been given, out of which three are alike in some manner and one is different. Select the number that is different.

Options:

A.

105

B.

348

C.

252

D.

231

Answer: B

Solution:

The logic followed here is:

All the given options except option (2), are divisible by 7.

Option (1) 105 → Divisible by 7.

Option (2) 348 → Not divisible by 7.

Option (3) 252 → Divisible by 7.

Option (4) 231 → Divisible by 7.

Hence, “348” is an odd one.

Question 31

One morning after sunrise, Vikram and Shailaish were standing in a park with their backs towards each other. Vikram's shadow fell exactly towards his left. In which direction was Shailaish facing?

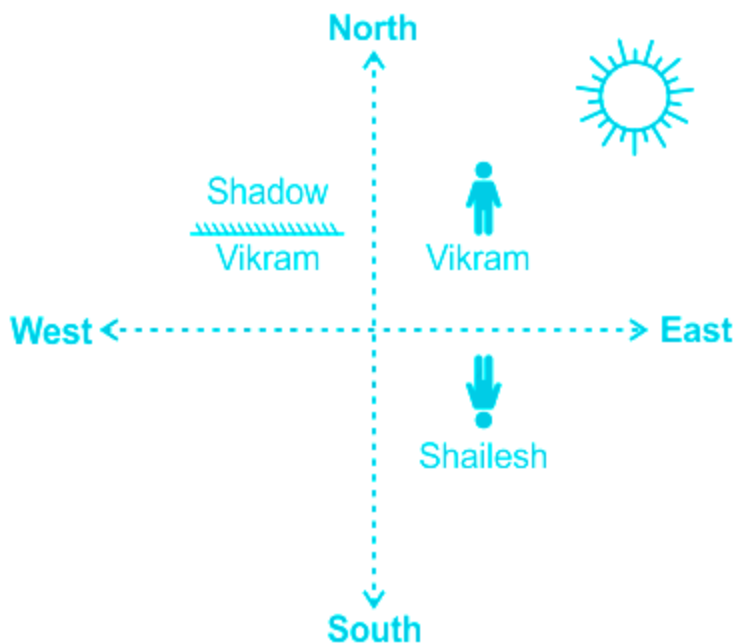
Options:

- A. North
- B. East
- C. South
- D. West

Answer: C

Solution:

The possible diagram according to statement is:



In this diagram, It is clearly shown that Shailaish is facing **South** direction.

Hence, "**South**" is the correct answer.

Additional Information

- When morning time, We will show the sun diagram in the east direction or right side because the sun rises in the east.
 - When Evening time, We will show the sun diagram in the west direction or left side because sunset in the west.
-

Question 32

Two bells ring at intervals of 63 seconds and 74 seconds. If they both ring at 10 O'clock in the morning together, after how many seconds will they ring together again?

Options:

A. 4482

B. 4662

C. 4552

D. 4772

Answer: B

Solution:

The logic follows here is:

Two bells ring at intervals of 63 seconds and 74 seconds.

Both the bells ring at 10 o'clock in the morning together

Now,

Both the bells will ring together again = LCM (63, 74)

According to the prime factorisation method,

3	6	3
3	2	1
7	0	7
	1	

2	74
37	37
	1

The factors of $63 = 3 \times 3 \times 7$

The factors of $74 =$

$$\Rightarrow \text{LCM}(63, 74) = 3 \times 3 \times 7 \times 2 \times 37$$

$$\Rightarrow \mathbf{4662}$$

Therefore, Both the bells will ring together again after 4662 seconds.

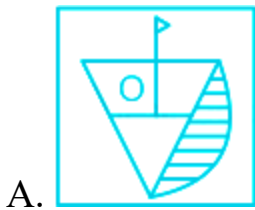
Hence, the correct answer is "**4662**".

Question 33

Which answer figure is the exact mirror image of the given question figure when mirror is held at PQ?



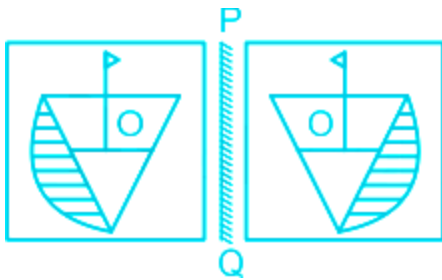
Options:



Answer: C

Solution:

In this question, We have to find the mirror image of the given image in the question, which is shown below-



Here, the mirror image is similar to Option (3).

Hence, **Option (3)** is correct.

Question 34

Select the figure that will replace the question mark (?) in the following figure series.



Options:



Answer: B

Solution:

The figure that will replace the question mark (?) in the following figure series is shown below:

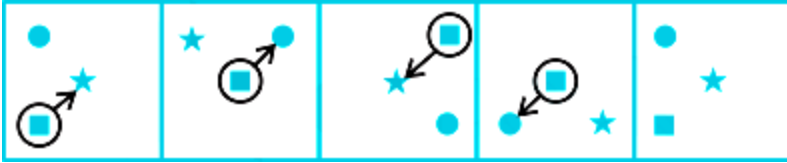
1)For '●'element :Shifting in the clockwise directionsubsequently.



2)For '★'element :Shifting diagonally to left corner upwards then rightcorner downwardsdirectionsubsequently.



3) For '■' element : Shifting diagonally to right corner upward then left corner downward directions subsequently.



Final series is :



Hence, 'option - (2)' is the correct answer.

Question 35

Six students are sitting around a circular table facing the centre. Salman is an immediate neighbour of both Kiran and Shwetha. Bala is sitting second right to Salman. Shwetha is sitting third to the left of Bala. Akshay is an immediate neighbour of both Bala and Nishitha. Who is the immediate neighbour of Nishitha and Salman?

Options:

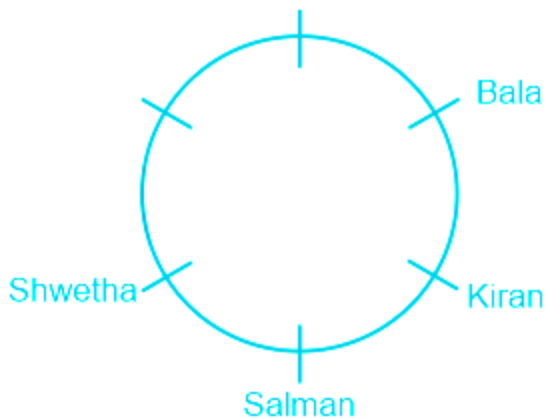
- A. Bala
- B. Kiran
- C. Shwetha
- D. Akshay

Answer: C

Solution:

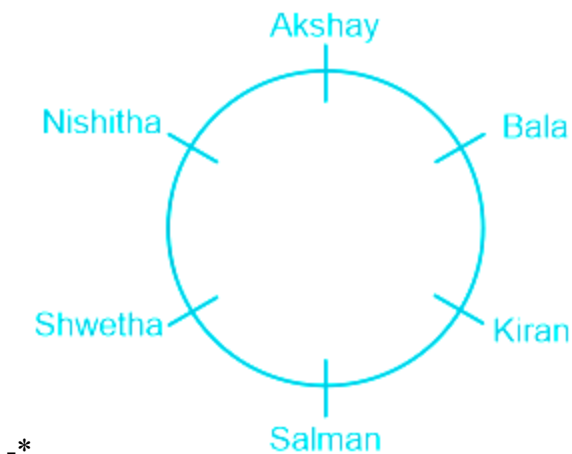
*-From the given information: /

Step 1: Salman is an immediate neighbour of both Kiran and Shwetha. Bala is sitting second right to Salman.



We place Salman in between Shwetha and Kiran and then place Bala second to the right of Salman.

Step 2: Shwetha is sitting third to the left of Bala. Akshay is an immediate neighbour of both Bala and Nishitha.



_*

We place Akshay next to Bala and then place Nishitha at the last spot left.

From the arrangement, we can see that **Shwetha is the immediate neighbour of Nishitha and Salman.**

Hence **the correct answer is option 3).**

Question 36

If Ammonia : Gas then Camphor : ?

A. Gas

B. Solid

C. Liquid

D. Semi-solid

Options:

A. A

B. C

C. B

D. D

Answer: C

Solution:

The relation is:

Ammonia: Gas $\rightarrow$ Ammonia is a compound of nitrogen and hydrogen with the formula NH_3 . Ammonia is a colourless gas with a characteristic pungent smell.

Ammonia is found in a gaseous state.

Similarly,

Camphor : ? $\rightarrow$ Camphor is a waxy, flammable, transparent solid with a strong aroma. It is a terpenoid with the chemical formula $\text{C}_{10}\text{H}_{16}\text{O}$. So, Camphor is found in solid-state.

Hence, '**solid**' is the correct answer.

Question 37

If A = 1, LION = 23 and CAMEL = 16 then what is the value of TIGER?

Options:

A. 36

B. 46

C. 32

D. 56

Answer: C

Solution:

A	B	C	D	E	F	G	H	I	J	K	L	M
1	2	3	4	5	6	7	8	9	10	11	12	13
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
14	15	16	17	18	19	20	21	22	23	24	25	26

A = 01

LION = L (1 + 2 = 3) + I (0 + 9 = 9) + O (1 + 5 = 6) + N (1 + 4 = 5) = 23

CAMEL = C (0 + 3) + A (0 + 1 = 1) + M (1 + 3 = 4) + E (0 + 5 = 5) + L (1 + 2 = 3) = 16

Similarly,

TIGER = T (2 + 0 = 2) + I (0 + 9 = 9) + G (0 + 7 = 7) + E (0 + 5 = 5) + R (1 + 8 = 9) = 32

Hence, the correct answer is 32.

Question 38

In the series next term will be
3, 11, 23, 39, 59,

Options:

A. 63

B. 73

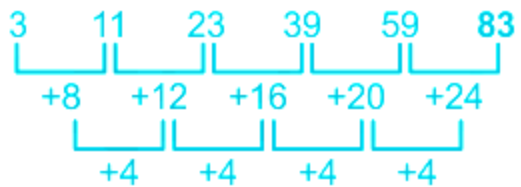
C. 83

D. 93

Answer: C

Solution:

The logic is:



Hence, '83' is the correct answer.

Question 39

In a certain code language, 'HYPOCRISY' is written as 'YPHOCIRYS', and 'IMPORTANT' is written as 'MPIORATTN'. How will 'INTEGRITY' be written in that language?

Options:

- A. NTIEGIRYT
- B. NTIEGRIYT
- C. NTIGEIRYT
- D. NITEGIRYT

Answer: A

Solution:

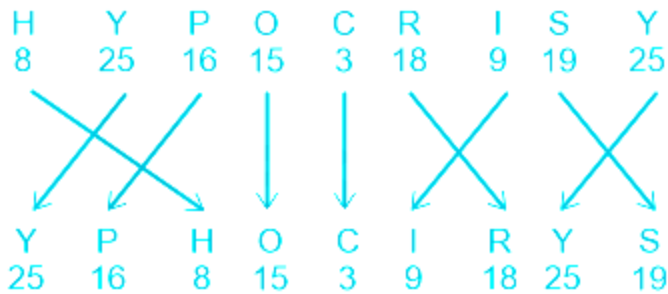
Alphabets	A	B	C	D	E	F	G	H	I	J	K	L	M
Positional value	1	2	3	4	5	6	7	8	9	10	11	12	13
Positional value	26	25	24	23	22	21	20	19	18	17	16	15	14
Alphabets	Z	Y	X	W	V	U	T	S	R	Q	P	O	N

The logic followed here is :

1. First letter - Shifted to the third position.
2. Second letter- Shifted to first position.
3. Third letter - Shifted to second position.
4. Fourth and fifth letter - No change
5. Sixth and seventh letter - Interchange position with each other.
6. Eighth and ninth letter -Interchange position with each other.

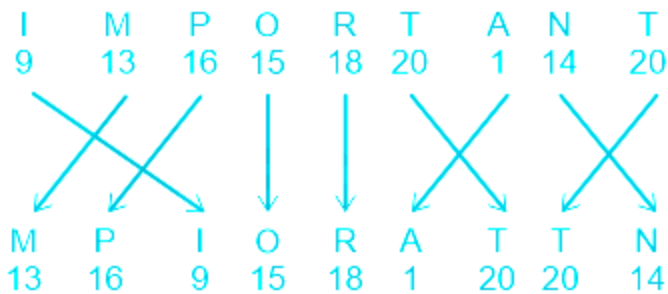
So,

- 'HYPOCRISY' is written as 'YPHOCIRYS'



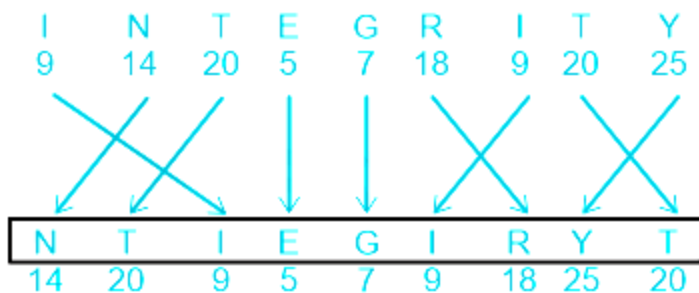
And,

- 'IMPORTANT' is written as 'MPIORATTN'



Similarly,

- 'INTEGRITY' is written as :

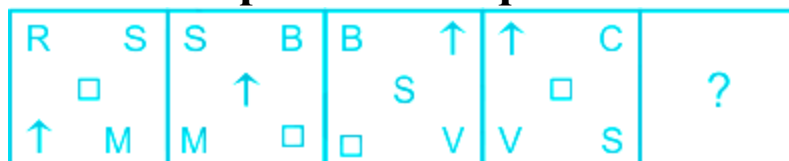


Hence, "NTIEGIRYT" is the correct answer.

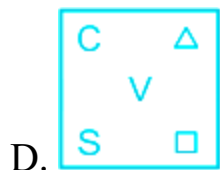
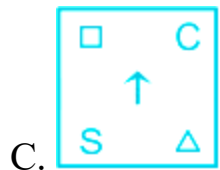
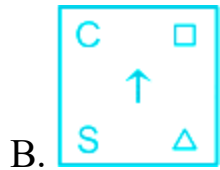
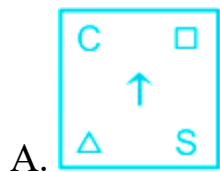
Question 40

Select the figure that will come in place of the question mark (?) in the

following figure series.



Options:



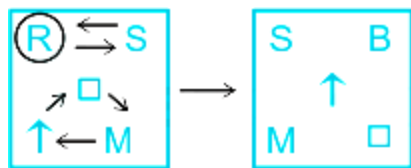
Answer: B

Solution:

The logic followed in the given figure series:

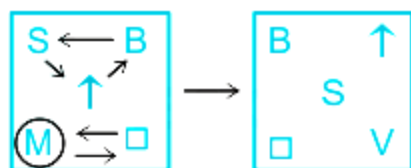
From Figure (1) to (2):

The symbols move as shown below and R changes into B.



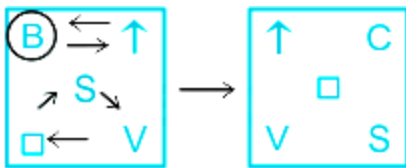
From Figure (2) to (3):

The symbols move as shown below and M changes into V.



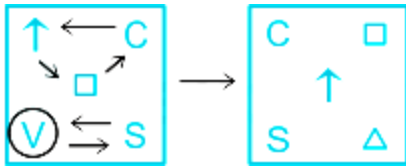
From Figure (3) to (4):

The symbols move as shown below and B changes into C.



From Figure (4) to Answer Figure (5):

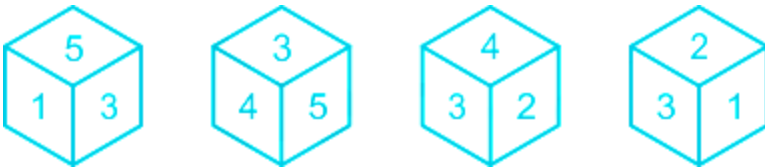
The symbols move as shown below and V changes into a triangle.



Hence, the correct answer is "Option (2)".

Question 41

Study the different positions of a dice given below. Which numbers will occur on the face opposite to 4?



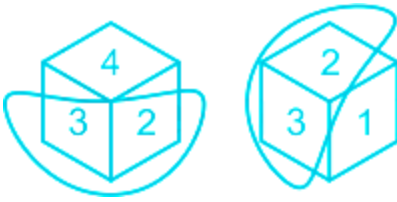
Options:

- A. 3
- B. 5
- C. 1
- D. 6

Answer: C

Solution:

Taking Dice position (3) and (4), as 2 and 3 are common;



The remaining number in Dice position (3) is 4.

The remaining number in Dice position (4) is 1.

So, 1 and 4 are opposite number pairs.

Hence, "1" is the correct answer.

Question 42

Study the given pattern carefully and select the number that can replace the question mark (?) in it

16	14	18
58	44	?
13	8	4

Options:

A. 44

B. 24

C. 36

D. 40

Answer: A

Solution:

The logic followed here is :-

In a column, **(First number + Last number) × 2 = Middle number.**

Column 1) :

16
58

13

$$= (16 + 13) \times 2$$

$$= 29 \times 2$$

$$= 58$$

Column 2) :

14
44
8

$$= (14 + 8) \times 2$$

$$= 22 \times 2$$

$$= 44$$

Column 3) :

18
?
4

$$= (18 + 4) \times 2$$

$$= 22 \times 2$$

$$= 44$$

Hence, the correct answer is "44".

Question 43

Simplify

$$2\frac{2}{3} \left[3\frac{1}{2} + \left\{ \frac{4}{5} - \left(3\frac{1}{2} \div \frac{14}{5} \right) \right\} \right]$$

Options:

A. $\frac{102}{15}$

B. $\frac{132}{15}$

C. $\frac{122}{15}$

D. $\frac{112}{15}$

Answer: C

Solution:

Formula used:

B	Brackets in order (), {}, []	ब्रैकेट (), {}, [] क्रम में
O	of	का
D	Division (÷)	विभाजन (÷)
M	Multiplication (×)	गुणा (×)
A	Addition (+)	जोड़ (+)
S	Subtraction (−)	घटाव (−)

Calculation:

By using the BODMAS rule

$$\Rightarrow \frac{8}{3} \left[\frac{7}{2} + \left\{ \frac{4}{5} - \left(\frac{7}{2} \times \frac{5}{14} \right) \right\} \right]$$

$$\Rightarrow \frac{8}{3} \left[\frac{7}{2} + \left\{ \frac{4}{5} - \frac{5}{4} \right\} \right]$$

$$\Rightarrow \frac{8}{3} \left[\frac{7}{2} - \frac{9}{20} \right]$$

$$\Rightarrow \frac{8}{3} \times \frac{61}{20}$$

$$\Rightarrow \frac{122}{15}$$

∴ The required value is $\frac{122}{15}$.

Question 44

A seller sells 2 articles, one of them for Rs 1200 at 20% profit and another one for Rs 1350 at 10% loss. Find the overall gain percent.

Options:

- A. 2 percent
- B. 1 percent
- C. 4 percent
- D. 3 percent

Answer: A

Solution:

Given

One article sold at 20% profit = Rs 1200

2nd article sold at 10% loss = Rs 1350

Formula Used

Cost price = $[\text{Selling price}/(100 + \text{profit})] \times 100$

Cost price = $[\text{Selling price}/(100 - \text{loss})] \times 100$

Calculation

Cost price of one article = $(1200/120) \times 100 = \text{Rs } 1000$

Profit = $1200 - 1000 = \text{Rs } 200$

Cost price of 2nd article = $(1350/90) \times 100 = \text{Rs } 1500$

Loss = $1500 - 1350 = 150$

So, overall Profit = Rs 50

Profit percentage overall = $(50/2500) \times 100 = 2\%$

$\therefore$ Gain percentage is 2%

Question 45

The average of 45 numbers is 150. Later it is found that a number 46 is wrongly written as 91, then find the correct average.

Options:

A. 151

B. 147

C. 149

D. 153

Answer: C

Solution:

Given:

The average of 45 data is 150

46 is wrongly written as 91

Concept used:

Average = Sum of total observations/Total number of observations

Calculation:

The total sum of all 45 number = $150 \times 45 = 6750$

Now, 46 is wrongly written as 91

The correct sum of data = $6750 - (91 - 46) = 6705$

Then, Correct average of the data = $6705/45 = 149$

$\therefore$ The correct average is 149

Shortcut Trick

Difference between wrong and actual numbers = $91 - 46 = 45$

As the actual number is less than the wrong number

So the average decreased by $45/45 = 1$

The correct average = $150 - 1 = 149$

$\therefore$ The correct average is 149

Question 46

30% of a number is 21 more than 15% of 120. What is the number?

Options:

- A. 95
- B. 110
- C. 125
- D. 130

Answer: D

Solution:

Calculation:

Let the number be A

$$30\% \text{ of } A = (15\% \text{ of } 120) + 21$$

$$\Rightarrow 0.3A = 18 + 21$$

$$\Rightarrow 0.3A = 39$$

$$\Rightarrow A = 130$$

$\therefore$ The number is 130

Question 47

In what time at 10% per annum simple interest a sum becomes 3 times of itself.

Options:

- A. 10 years
- B. 15 years
- C. 20 years
- D. 25 years

Answer: C

Solution:

Given:

$$\text{Rate} = 10\%$$

$$\text{Amount} = 3 \times \text{Principal}$$

Formula Used:

$$\text{S.I} = \text{PRT}/100$$

$$\text{A} = \text{P} + \text{S.I}$$

Where, S.I = Simple Interest

P = Principal, R = Rate,

T = Time, A = Amount

Calculation:

According to the question

$$\text{A} = 3\text{P}$$

$$\text{P} + \text{S.I} = 3\text{P}$$

$$\Rightarrow \text{S.I} = 2\text{P}$$

$$\Rightarrow (\text{P} \times 10 \times \text{T})/100 = 2\text{P}$$

$$\Rightarrow \text{PT}/10 = 2\text{P}$$

$$\Rightarrow \text{T}/10 = 2$$

$$\Rightarrow \text{T} = 20 \text{ years}$$

$\therefore$ The period of time is 20 years.

Question 48

Two trains are running on opposite tracks between stations A and B. After crossing each other they take 4 hr and 9 hr respectively to reach their destination. If speed of first train is 54 kmph, find the speed of second train.

Options:

- A. 18 kmph
- B. 36 kmph
- C. 44 kmph
- D. 28 kmph

Answer: B

Solution:**Given:**

Two trains are running on opposite tracks between stations A and B.

After crossing each other they take 4 hr and 9 hr respectively to reach their destination.

Speed of first train is 54 kmph.

Formula used:

After crossing each other, if time taken by 2 trains is T_1 and T_2 resp. then, $S_1/S_2 = \sqrt{T_2}/\sqrt{T_1}$

where, S_1 and S_2 are speeds of first and second train respectively

Calculation:

We have, $T_1 = 4\text{hr}$, $T_2 = 9\text{hr}$, $S_1 = 54\text{ kmph}$

$$\Rightarrow 54/S_2 = \sqrt{9/4} = 3/2$$

$$\Rightarrow S_2 = 54 \times 2 \times 1/3 = 36\text{ kmph}$$

$$\Rightarrow \text{Speed of second train} = 36\text{ kmph}$$

**Alternate Method**

Let the speed of the second train be ' x ' kmph

Also, time taken to cross each other $= \sqrt{(T_1 \times T_2)} = \sqrt{(9 \times 4)} = 6\text{ hrs}$

$$\text{Total distance} = 54 \times 6 + x \times 6 = x \times 9 + 54 \times 4$$

$$\Rightarrow 9x - 3x = 54 \times (6 - 2)$$

$$\Rightarrow 6x = 216$$

$$\Rightarrow x = 36\text{ kmph} = \text{Speed of second train}$$

Question 49

The sum of two numbers is 64 and their ratio is 7 : 9. The sum of digits of these numbers is:

Options:

A. 12

B. 13

C. 19

D. 16

Answer: C

Solution:

The sum of two numbers is 64 and their ratio is 7 : 9.

Let numbers be $7a$ and $9a$, where 'a' is any constant

$$\Rightarrow 7a + 9a = 64$$

$$\Rightarrow 16a = 64$$

$$\Rightarrow a = 4$$

$\therefore$ Numbers are

$$7a = 28$$

$$9a = 36$$

$$\therefore \text{Sum of digits of these numbers} = 2 + 8 + 3 + 6 = 19$$

Question 50

Length of a rectangle is twice its breadth. If the perimeter of the rectangle is 78 m, then what will be the area of the rectangle?

Options:

A. 300 m^2

B. 338 m^2

C. 169 m^2

D. 507 m^2

Answer: B

Solution:**Given:**

Length of rectangle = $2 \times \text{Breadth}$

Perimeter = 78 m

Formula used:

$$\text{Perimeter} = 2 \times (L + B)$$

$$\text{Area} = L \times B$$

Where L is length and B is the breadth

Calculation:

Let the breadth of the rectangle be x m and the length of the rectangle be 2x m

$$\text{Perimeter} = 2 \times (2x + x) = 6x$$

$$\Rightarrow 78 = 6x$$

$$\Rightarrow x = 13 \text{ m}$$

$$\text{Length} = 2 \times 13 = 26 \text{ m and Breadth} = 13 \text{ m}$$

$$\text{Area} = L \times B$$

$$\Rightarrow \text{Area} = 13 \times 26$$

$\therefore$ Area of rectangle is 338 m^2
